Santulan: a research prototype for clinician-led diabetes dose review

Architecture and data flow. Each part is labelled by its status. The clinician decides; the software never changes a dose.

implemented / evaluated · illustrative / experimental · example / proposed

INPUTS

Clinic record (EHR)

Age, HbA1c, eGFR, C-peptide, BMI
Medicines and dose records
CYP2C9, G6PD where tested (else unknown)

EXAMPLE SHAPES
FHIR R4 modelled on the ABDM guide; not validated

CGM and daily logs

CGM every 15 min; 5 days calibrate the twin
Fingersticks, meals, insulin and tablet doses

REAL DATA
ShanghaiT2DM: 100 adults with Type 2 diabetes, 109 sensor periods (CC BY 4.0)

Smartwatch

Heart rate, steps, sleep

DISPLAY ONLY
Synthetic Health Connect shapes; not used by any model

THE TWIN, ONE PER PATIENT

1. Calibration (fixed 5 days) — IMPLEMENTED

Sensor offset from calibration fingersticks only, then frozen; meal fill and baseline drug flags from calibration records

2. Physiology core (fitted per person) — IMPLEMENTED

Linearised Bergman minimal model with insulin, sulfonylurea, GLP-1 and meal curves; sign-constrained least squares

Lowers 2-hour error by 2.2 mg/dL (1.1 in the full hybrid), held-out patients

3. Learning layer (LightGBM, across patients) — EVALUATED

CGM history, logged inputs, clinical record, core forecasts; tested only on patients excluded from its training (5 patient folds)

The clinical record did not improve forecast accuracy

4. Review rules and safety notes — RULES

GLP-1 start, weight loss, planned fast, eGFR below 60; G6PD deficiency, CYP2C9 variant, age 65 and over

Stated rules with label notes, judged as of a date; not validated

OUTPUTS TO THE CLINICIAN

Glucose forecast, 30 / 60 / 120 min

RMSE 13.2 / 23.0 / 32.6 mg/dL

vs 16.6 / 27.6 / 40.7 for no change

Low alert (simple rule)

Corrected reading at or below its threshold

43 of 53 lows at 0.5 alerts a day; models not better

Dose scenarios

Options side by side on the person's core

Illustrative; assumptions and sensitivity shown

Sensor-off estimation

RMSE 29.6 mg/dL from doses, meals, fingersticks

Experimental: no low detection; at most 8.0 days tested

Clinician dashboard (Streamlit)

Live replay of a held-out real patient, review list, scenarios, evidence. The clinician decides.

The clinician's decision changes the patient; a new sensor period recalibrates the twin (no online refitting)

LIFECYCLE

Calibrate — CGM fortnight, fingersticks and labs fit offset and core

Watch, with the CGM on — Forecasts and the simple alert; live replay shows this loop

Review and discuss — Rules flag the record; options compared under stated assumptions

Re-sense (proposed) — Quarterly sensor periods; cost and safety need a prospective pilot

Evaluation protocol v2 (docs/evaluation_protocol.md)

Calibration-only personal state; offset prior, models and alert thresholds fitted on training patients only; one alert policy and one-to-one scoring; the live replay matches the evaluation's stored predictions (tested). Stack: Python, pandas, SciPy, LightGBM, Streamlit, Plotly. Apache-2.0.